The *Plasmodium vivax* PIR Family Across Three High-Quality Reference Assemblies

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Abstract

We characterize the PIR (*Plasmodium* interspersed repeat) gene family in three modern high-quality reference assemblies of *Plasmodium vivax*: PvP01 (Papua, Indonesia), PvW1 (Thailand), and PvPAM (Peruvian Amazon). The 2,664 PIR proteins identified by Pfam HMM (PF05795) resolve into 73 structural classes by Foldseek + ProstT5 clustering; one class (FS_001) accounts for 46% of the family by itself, and ten classes together hold 95% of all PIRs. Twenty-one of the 73 classes are strictly universal across the three continents and account for 97% of the family by membership. PIRs occupy all 14 main chromosomes of each assembly in subtelomeric blocks where members of different structural classes are interleaved, and the per-class spatial signature is highly conserved between assemblies (Spearman $\rho \approx 0.91$). Of 194 reciprocal best hits within the dominant structural class between the two long-read assemblies (PvW1 and PvPAM), only 11 survive microsynteny verification as high-confidence 1-to-1 orthologs. Six-frame translation of pseudogenic transcripts followed by PF05795 HMM search identifies 75 and 70 VIR-family pseudogenes in PvP01 and PvPAM respectively (PvW1 release does not annotate pseudogenes). Within the dominant structural class, sequence diversity is high but not partitioned by geographic origin, consistent with a family whose diversification predates the modern *P. vivax* lineages and is maintained by subtelomeric recombination across continents.

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0.1 Introduction

The PIR (*Plasmodium* interspersed repeat) family is the largest multigene family in *Plasmodium vivax* and one of the largest in any eukaryotic genome. Its members encode variant surface antigens that are exported to the surface of infected reticulocytes, where they appear to participate

in antigenic variation, cytoadherence to endothelial receptors, and the establishment of chronic asymptomatic infections. In *P. vivax* the family is called *vir* (for "*P. vivax* variant"), and is part of the broader PIR superfamily that spans every sequenced *Plasmodium* species except *P. falciparum*. The defining structural feature, established by the crystal structure of a *P. chabaudi* PIR ectodomain (Harrison & Higgins, 2020, PDB 6ZYV), is a small, mostly alpha-helical fold that is preserved across the entire superfamily despite extensive sequence divergence between individual members. This conserved fold is captured at the sequence-profile level by a hidden Markov model in the Pfam database (Mistry et al., 2021, *Nucleic Acids Res.* 49:D412) with accession PF05795, which is the operational definition of a PIR/VIR protein used throughout this analysis: any annotated protein in a *P. vivax* proteome that matches the PF05795 profile with a significant E-value, found with `hmmsearch` from HMMER 3 (Eddy, 2011, *PLoS Comput. Biol.* 7:e1002195). This structural conservation is the central reason why characterizing the family at scale requires structure-aware methods: pairwise sequence identity within the family falls well below the so-called twilight zone of 20-25%, where standard sequence-based homology methods break down.

Until recently, the *vir* family in *P. vivax* was known almost exclusively from the Salvador-I reference genome (Carlton et al., 2008), in which 346 *vir* genes were annotated and organized into twelve subfamilies labelled A through L on the basis of pairwise sequence similarity. The Salvador-I assembly was produced from Sanger sequencing of a monkey-adapted strain and is known to be highly fragmented in subtelomeric regions, where the bulk of the *vir* repertoire resides. In the past decade, three considerably more contiguous reference genomes from geographically distant *P. vivax* lineages have become available: PvP01 from Papua, Indonesia (Auburn et al., 2016), assembled from very high-depth Illumina sequencing with manual curation of the subtelomeres; PvW1 from Thailand (Minassian et al., 2021), a hybrid Illumina + PacBio assembly produced for controlled human malaria infection studies; and PvPAM from the Peruvian Amazon (Vaughan et al., 2023), a PacBio-only assembly produced from a low-input patient sample. These three references together span Asia, Southeast Asia, and the Americas, and each resolves the subtelomeric *vir* repertoire to a different degree.

This section characterizes the *P. vivax* PIR family across these three modern assemblies, using a structure-aware computational pipeline based on Foldseek + ProstT5. Foldseek (van Kempen et al., 2024, *Nat. Biotechnol.* 42:243) is a structural homology search tool that encodes the local backbone geometry of every residue in a protein structure as one of twenty letters in the so-called 3Di structural alphabet, and then aligns proteins using both the amino acid alphabet and the 3Di alphabet simultaneously. ProstT5 (Heinzinger et al., 2024, *NAR Genom. Bioinform.* 6:lqae150) is a protein language model trained to predict the 3Di alphabet directly from amino acid sequence, so the combination allows structural clustering of large protein sets without ever computing three-dimensional structures and FoldMason (Gilchrist et al., 2026, *Science*, doi:10.1126/science.ads6733) and IQ-TREE 3 (Trifinopoulos et al., 2026; the 2020 paper Minh et al., *Mol. Biol. Evol.* 37:1530, describes IQ-TREE 2 which is the methodological reference). FoldMason performs progressive multiple structural alignment using a guide tree built from all-versus-all 3Di+AA pairwise comparisons, producing aligned amino acid and 3Di sequences in lockstep. IQ-TREE 3 is a maximum-likelihood phylogenetic inference engine with built-in substitution model selection (ModelFinder; Kalyaanamoorthy et al., 2017, *Nat. Methods* 14:587), efficient ultrafast bootstrap (UFBoot; Hoang et al., 2018, *Mol. Biol. Evol.* 35:518) and the SH-aLRT branch test (Anisimova & Gascuel, 2006, *Syst. Biol.* 55:539).. The goal is descriptive: to provide an organized account of how many PIR proteins exist in each assembly, how they group into structural classes, how those classes are distributed across the genome, which members can be confidently mapped between assemblies as 1-to-1 orthologs, and what the within-class diversity looks like.

The Salvador-I reference is referred to throughout as the historical baseline but is not re-analyzed at the protein level here, because its subtelomeric fragmentation makes per-protein structural and spatial characterization unreliable – a comparison that would be useful but requires re-assembly work beyond the scope of this characterization.

0.2 Assembly resources and PIR census

The three assemblies differ substantially in sequencing technology, contiguity, and consequently in how completely they resolve subtelomeric *vir* clusters. Table 1 summarizes the relevant properties as they affect this analysis. Per-protein characterization in this section is based on the predicted functional proteome of each assembly as released through PlasmoDB release 71 (Amos et al., 2022, *Nucleic Acids Res.* 50:D898; PlasmoDB is the *Plasmodium* genome database within the VEuPathDB resource, providing curated genome annotations, gene ontology mappings, and orthology calls for malaria parasite genomes). Pseudogenes are analyzed separately. The PvP01 and PvPAM gene-level GFF annotations from PlasmoDB release 68 include 185 and 322 pseudogene records respectively; the PvW1 release does not include any pseudogene calls (this is an annotation gap inherited from Minassian et al., 2021, not a biological claim). Extracting the nucleotide sequences of the pseudogenic transcripts, translating in all six reading frames, and searching the translations with the PF05795 HMM at $E < 10^{-3}$ identifies 75 of the 185 PvP01 pseudogenes and 70 of the 322 PvPAM pseudogenes as VIR-family pseudogenes. The PvP01 set is a perfect subset of the 119 PvP01 pseudogenes that PlasmoDB product-labels as PIR/VIR, with the 44 product-labeled-but-HMM-missed cases representing severely fragmented pseudogenes that have lost detectable PF05795 signal. The PvPAM pseudogenes carry no product annotation in the GFF, so the HMM-based count of 70 is the only available characterization for that genome.

Table 1. Reference assemblies characterized in this study.

Property	PvP01	PvW1	PvPAM
Geographic origin	Papua, Indonesia	Thailand	Peruvian Amazon
Reference accession (NCBI)	GCA_900093555.2	GCA_914969965.1	GCA_949152365.1
Sequencing technology	Illumina (high depth) + manual curation	PacBio + Illumina (hybrid)	PacBio (Ultra-Low Input)
Year	2016	2021	2023
Total annotated proteins (PlasmoDB v71)	6,523	6,075	6,462
VIR proteins identified by Pfam HMM (PF05795)	931	748	985
Contigs bearing at least one VIR	130	17	26
Largest contig	3.2 Mb	3.7 Mb	3.3 Mb
VIRs on the 14 largest contigs (“main chromosomes”)	347 (37%)	634 (85%)	735 (75%)
VIRs on small subtelomeric contigs	584 (63%)	114 (15%)	250 (25%)

Property	PvP01	PvW1	PvPAM
Combined main-chromosome length	23.5 Mb	27.7 Mb	27.4 Mb
Pseudogenes (any family) annotated in PlasmoDB release 68 GFF	185	0 (not annotated)	322
VIR-family pseudogenes (PF05795 hits on 6-frame translation, E $< 10^{-3}$)	75	not determinable	70

The most important practical consequence of this table is that PvP01 places only 37% of its identified VIR proteins on main chromosomes. The remaining 584 PvP01 VIRs sit on small subtelomeric contigs (<200 kb each) that the original Illumina assembly was unable to scaffold into the main chromosomes. PvW1 and PvPAM, both produced with PacBio long reads, retain 75-85% of their VIR repertoire on the main chromosomes. This affects every spatial statistic that follows: PvP01's apparent subtelomeric distribution and chromosome-to-chromosome distribution metrics are confounded by its high contig count, and the long-read assemblies should be considered authoritative for those measurements. Sequence- and structure-level characterization (clustering, phylogeny, orthology) is largely unaffected, because those analyses operate on protein sequences and not on chromosomal context.

A direct comparison to the historical Salvador-I reference is not attempted here. The Sal-I assembly contains 2,747 scaffolds and lacks the contiguity needed for the spatial and orthology analyses below. The Lopez-Perez et al. (2013) re-analysis of Sal-I revised the original 346 *vir* gene count downward (concluding that subfamilies A, D, and H were assembly artifacts rather than true *vir* genes), and modern long-read assemblies have shown that the true *P. vivax* VIR repertoire contains over a thousand functional copies per genome. Salvador-I therefore serves as historical reference but not as a peer dataset.

0.3 Structural classes

A central characterization question for any large multigene family is "how many distinct kinds of members does it contain?" Because *vir* proteins are too divergent in sequence for conventional clustering to give a meaningful answer (median pairwise identity within FS_001 is around 30-40%), we used Foldseek's `easy-cluster` workflow with ProstT5 encoding. The workflow runs an all-versus-all structural homology search of the input proteins against themselves and then groups them with a greedy set-cover algorithm (Steinegger & Söding, 2018, *Nat. Commun.* 9:2542) on the resulting graph of pairwise similarities. The 3Di alphabet representation respects the conserved PIR fold even where amino acid identity has been lost. This approach respects the conserved PIR fold even where sequence similarity has been lost.

The 2,664 HMM-identified VIRs from the three assemblies were clustered together. Foldseek produced 73 clusters containing three or more proteins, which we labeled FS_001 through FS_073

in decreasing order of size. The size distribution is heavily skewed: FS_001 alone contains 1,197 proteins, or 46% of the entire family. FS_002 through FS_005 add another 912 proteins (35% of the family). The remaining 68 clusters together hold the final 19% of the family, mostly as small clusters of two to thirty members.

Table 2. Top structural classes and their per-genome composition.

Cluster	Total	PvP01	PvW1	PvPAM	Fraction of total VIR family
FS_001	1,197	400	349	448	46.2%
FS_002	385	126	101	158	14.9%
FS_003	248	77	78	93	9.6%
FS_004	143	53	41	49	5.5%
FS_005	136	48	37	51	5.2%
FS_006	96	28	33	35	3.7%
FS_007	85	39	20	26	3.3%
FS_008	74	31	18	25	2.9%
FS_009	53	20	16	17	2.0%
FS_010	36	15	7	14	1.4%
(top 10 subtotal)	2,453	837	700	916	94.6%
FS_011-FS_073	139	(rest)	(rest)	(rest)	5.4%

A comparison to the historical A-L subfamily classification of Salvador-I is informative but only partially possible: 89 of the 146 sequence-only subfamilies that we generated as a baseline (Plan A) collapse into FS_001 alone. The baseline was produced with MMseqs2 (Steinegger & Söding, 2017, *Nat. Biotechnol.* 35:1026), a fast sensitive protein sequence search tool, to compute an all-versus-all bitscore matrix, followed by the Markov Clustering Algorithm (MCL; van Dongen, 2008, *SIAM J. Matrix Anal. Appl.* 30:121), a graph-based clustering algorithm that groups proteins by simulating random walks on the similarity network with inflation parameter 2.0, indicating that several of the historical subfamilies that Lopez-Perez et al. (2013) treated as distinct are members of a single structural class. A full mapping of the original A-L assignments to our FS clusters would require running the Sal-I VIR sequences through the same clustering pipeline and is left as a follow-up.

The cross-assembly composition of the structural classes is informative on its own. Twenty-one of the 73 clusters strictly contain members from all three reference genomes. Three more clusters contain members from PvP01 and PvPAM only, one from PvW1 and PvPAM only, and one from PvP01 and PvW1 only. The remaining 47 clusters appear in only a single genome each, but together account for only 52 proteins (2.0% of the family) and are almost all singletons or pairs of closely related sequences. Weighted by membership, 2,523 of the 2,592 clustered proteins (97.3%) fall into the 21 strictly universal clusters. In other words, the structural classification of the *P. vivax* VIR family is essentially universal: a small number of large structural classes account for the bulk of the family in every assembly, and the apparent assembly-specific clusters reflect rare variants rather than independently-evolved structural classes.

0.4 Genomic organization

The protein-level census above is independent of where the genes live in the genome. To characterize the spatial organization of the family across the three assemblies, we mapped each VIR to its chromosomal coordinates, restricted attention to the 14 largest contigs per genome (the "main chromosomes"), and quantified subtelomeric concentration, per-cluster chromosomal distribution, and conservation of distribution across assemblies.

The basic spatial fact is that VIRs are concentrated at chromosome ends. Across all three assemblies, 64.5% of main-chromosome VIRs lie in the outer 10% of their chromosome by length, against a uniform-distribution expectation of 20%. Per-assembly enrichment is strongest in PvP01 (80% of main-chromosome VIRs in the outer 10%) and progressively weaker in PvW1 (64%) and PvPAM (57%). The PvP01 number is partly an artifact of its smaller main-chromosome footprint (only 347 VIRs on 23.5 Mb of main chromosome, vs 634/27.7 Mb for PvW1 and 735/27.4 Mb for PvPAM): the densely-VIR-bearing subtelomeric blocks that PvW1 and PvPAM place inside their main chromosomes are partially split off onto small contigs in PvP01.

Within the subtelomeric blocks, members of the same structural class do not form tandem arrays. Permutation tests for tandem clustering of same-cluster members against random label shuffles return only 2 significant results out of 32 (cluster, genome) tests at $p < 0.05$, consistent with chance under multiple testing. However, when we examine all VIR-VIR pairs within 50 kb of each other on the same chromosome, the fraction in which both members belong to the same structural class is consistently higher than expected: 35% / 31% / 32% observed in PvP01 / PvW1 / PvPAM against expectations of 20% / 24% / 23% under random assignment of clusters to subtelomeric positions. So structural classes co-occupy similar subtelomeric neighborhoods (with a 1.3-1.7-fold enrichment), but they are interleaved with members of other classes rather than forming pure tandem blocks.

A striking cross-assembly observation is that the per-cluster chromosomal footprint is highly conserved between genomes. The number of main chromosomes occupied by each universal cluster correlates with Spearman ρ (a rank-based correlation coefficient that captures monotonic association without assuming linearity) ≈ 0.91 between all three pairs of genomes (PvP01 vs PvW1, PvP01 vs PvPAM, PvW1 vs PvPAM). A cluster that lives on many chromosomes in PvP01 lives on many in PvW1 and PvPAM as well, and vice versa. FS_001, the dominant class, is present on all 14 main chromosomes in all three genomes, with 137 / 284 / 320 members on main chromosomes in PvP01 / PvW1 / PvPAM. The relative-position density of FS_001 members along chromosomes is essentially identical to that of all other VIRs taken together and is consistent across assemblies.

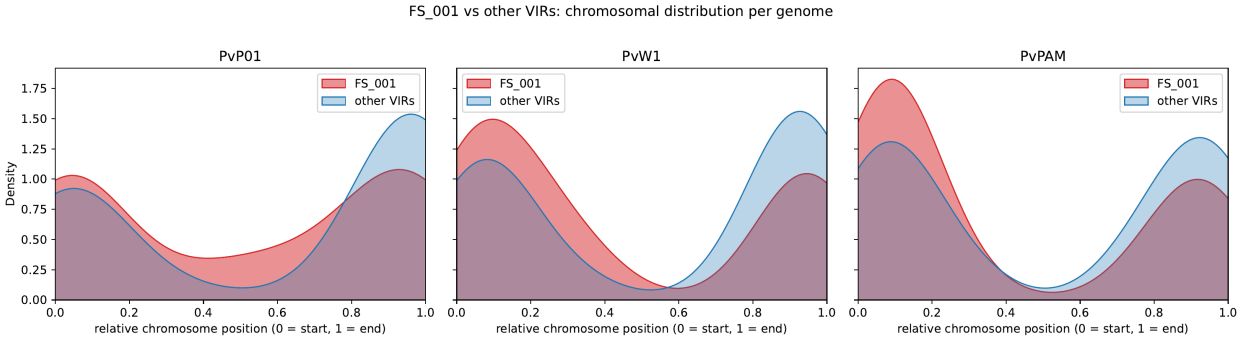


Figure 1: Relative-position density of FS_001 (red) and other VIR (blue) genes along main chromosomes, per assembly. Position is expressed as fraction of chromosome length.

The structural and spatial signatures of the family, in other words, are conserved features of *P. vivax*: they are not properties of any individual lineage.

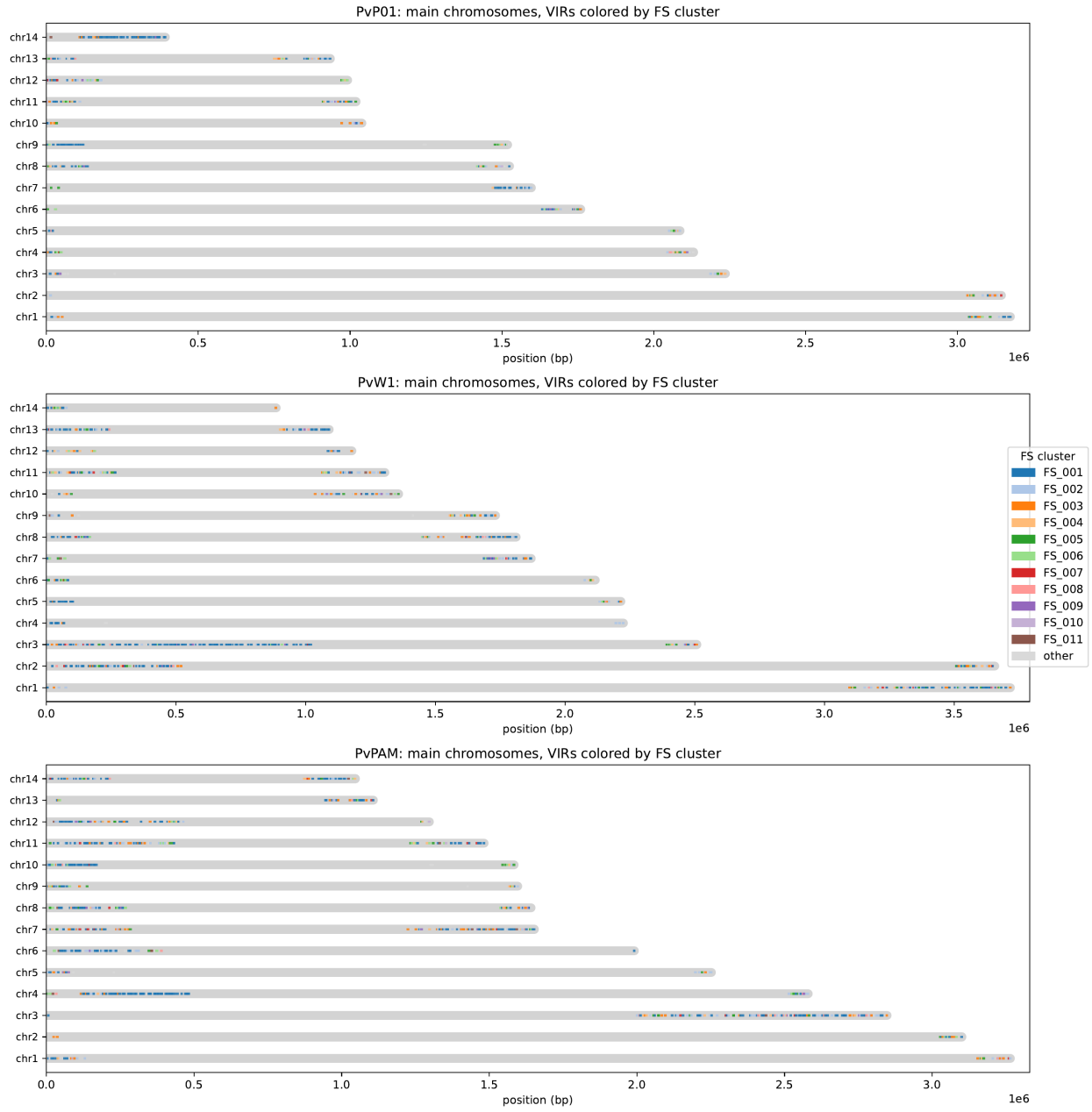


Figure 2: Chromosomal landscape of VIR genes across the 14 main chromosomes of each assembly. Each tick mark is a VIR gene, colored by Foldseek structural class (top 10 universal classes colored distinctly; remaining classes shown in gray).

0.5 Cross-assembly orthology of VIR genes

The fine-grained question that follows from these results is whether individual VIR genes can be confidently mapped 1-to-1 across the three assemblies, and how often that succeeds. We approached this between the two long-read assemblies (PvW1 and PvPAM) because their subtelomeric contiguity is sufficient to support microsynteny analysis; PvP01 was excluded because its fragmentation into many small subtelomeric contigs destroys the gene-order context needed for

microsynteny verification.

Reciprocal best hits (RBH; the standard operational definition of potential orthologs between two genomes, where two proteins A in genome 1 and B in genome 2 are reciprocal best hits if B is the best match for A in genome 2 and A is the best match for B in genome 1) using MMseqs2 within FS_001 recover 194 PvW1 vs PvPAM pairs with $E < 10^{-5}$, identity ≥ 30%, and mutual coverage ≥ 50%. To distinguish true 1-to-1 orthologs from paralog-best-hits we cross-checked each pair against two additional evidence layers. First, the housekeeping-protein RBH backbone of 5,120 non-VIR ortholog pairs between PvW1 and PvPAM, asking whether each FS_001 pair's flanking non-VIR neighbors are themselves housekeeping orthologs (microsynteny score 0-4). Microsynteny refers to the conservation of local gene order across genomes (Sonnhammer & Koonin, 2002, *Trends Genet.* 18:619): if two genes are true orthologs descended from a single ancestral gene, their immediate genomic neighbors should also be 1-to-1 orthologs more often than if the apparent orthology had resulted from independent duplications. Second, the FS_001 phylogeny from Plan C Q2, asking whether the pair forms reciprocally nearest-neighbor (sister) leaves on the AA tree (computed with Biopython's `Bio.Phylo` module, Cock et al., 2009, *Bioinformatics* 25:1422).

The microsynteny result is the most informative: 139 of the 194 RBH pairs (72%) have zero matching flanking orthologs, meaning the local gene neighborhoods around them have been completely rearranged between PvW1 and PvPAM. Only 11 RBH pairs (6%) have three or four of four flanking neighbors preserved as housekeeping orthologs. These 11 high-confidence ortholog candidates also have substantially higher sequence identity (median 87%, range 70-99.7%) than the bulk of the RBH set (median 74.5%), and several appear in clusters of consecutive gene IDs in both genomes (e.g., `CAG9483870` / `CAG9483878` / `CAG9483882` in PvW1 map to `CAI7717859` / `CAI7717862` / `CAI7717863` in PvPAM), suggesting they sit in the rare subtelomeric neighborhoods where local gene order is preserved between Thailand and Peru. These 11 pairs are the strongest within-FS_001 1-to-1 ortholog candidates available between PvW1 and PvPAM and are documented in `data/genomes/alignments/ortho/orthologs_high_confidence.tsv`.

The methodological implication is concrete and useful for downstream work on the family: reciprocal best hits alone do not establish orthology for subtelomeric VIR genes, because the subtelomeric environment is dynamic enough that paralogs frequently outscore the true ortholog. Microsynteny verification against a housekeeping-protein RBH backbone is a necessary additional filter.

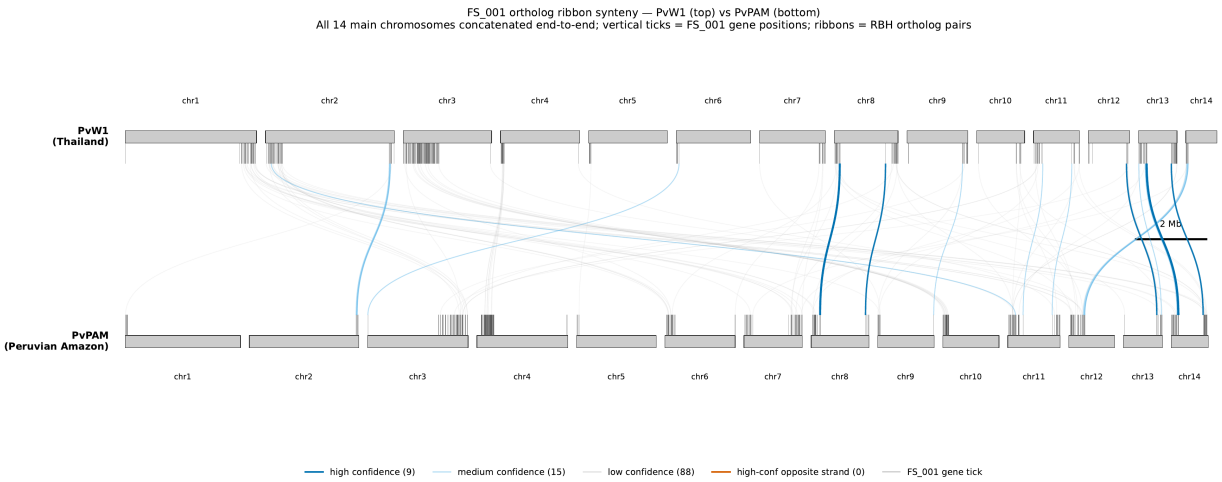


Figure 3: Ribbon synteny plot of FS_001 reciprocal best hits between PvW1 (top) and PvPAM (bottom). Ribbons colored by confidence: dark blue = 11 high-confidence orthologs; light blue = 26 medium-confidence; gray = 157 low-confidence.

For a typical biological question that needs orthologs (e.g., “is this PvW1 gene the equivalent of that PvPAM gene?”) the operational answer is that high-confidence orthology is available for roughly 6% of the FS_001 RBH set, or about 11 genes out of the ~349 PvW1 FS_001 members.

For the housekeeping (non-VIR) backbone, the situation is completely different: 5,120 RBH pairs between PvW1 and PvPAM at high identity (median 99.8%), confirming that the two lineages are evolutionarily close and that the difficulty of VIR orthology is a property of the family, not of the assemblies.

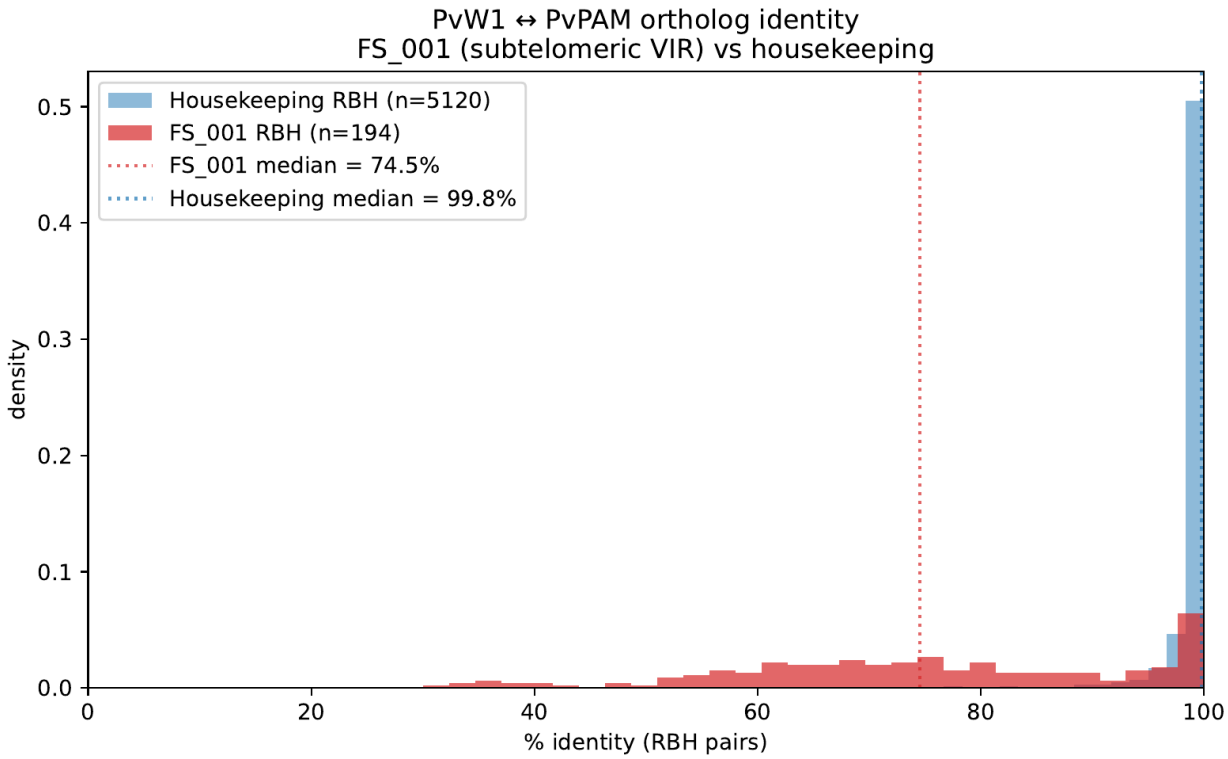


Figure 4: Pairwise sequence identity of reciprocal best hits between PvW1 and PvPAM. Housekeeping (non-VIR) ortholog set in blue (n=5,120); FS_001 RBH pairs in red (n=194).

0.6 Within-class diversity

For the dominant structural class FS_001 (1,197 proteins, 46% of the family), we examined the within-class sequence diversity directly. A 50% sequence-identity subsampling of FS_001 with MMseqs2 `easy-cluster` (the same tool used for the Plan A baseline, here run with `--min-seq-id 0.5 -c 0.8` to pick a representative for each group of sequences sharing >=50% identity over >=80% mutual coverage) retained 338 representative sequences (118 PvP01, 110 PvW1, 110 PvPAM, a balanced cross-continental sample). Maximum-likelihood phylogenies of these 338 representatives were inferred with IQ-TREE 3 on (a) the AA alignment alone (ModelFinder selected VT+F+R6, i.e. the VT amino acid exchangeability matrix of Müller & Vingron, 2000, *J. Comput. Biol.* 7:761, with empirical state frequencies and a 6-category FreeRate model of among-site rate variation); (b) the 3Di alignment using the Garg & Hochberg 2025 substitution matrix (*Mol. Biol. Evol.* 42:msaf124), the first general substitution model published for the 3Di alphabet, in NEXUS format from the `jonasmehl1/structural_phylogenetics` repository; and (c) a partitioned AA + 3Di analysis. Each tree was inferred with 1000 ultrafast bootstrap replicates (UFBoot; Hoang et al., 2018, *Mol. Biol. Evol.* 35:518, a fast resampling procedure for branch support) and 1000 SH-aLRT replicates (Anisimova & Gascuel, 2006, *Syst. Biol.* 55:539, an approximate likelihood-ratio test that complements UFBoot).

The within-FS_001 sequence diversity is high and not partitioned by geographic origin. Tips from the three genomes are evolutionarily interleaved across all three trees: the median parsimony score for "genome of origin" mapped onto each tree is 160 transitions, identical to the median under 1000 random shuffles of the tip labels, and far above the minimum-possible value of 2 (correspond-

ing to a single monophyletic clade per genome). Counting internal nodes whose descendants all came from a single genome gave 34 to 37 out of 336 internal nodes (10-11%), no more than expected by chance for three states distributed evenly across tips. The phylogenetic signature is consistent with what would be expected if the sequence diversity within FS_001 had been generated before the geographic separation of the three *P. vivax* lineages represented here, and had then been maintained subsequently by recombination and gene conversion within and between subtelomeric *vir* loci that operates faster than geographic isolation can partition the family.

A family-level phylogeny on the 73 Foldseek cluster representatives (Plan C Q1) did not resolve deep relationships between the structural classes cleanly, even after switching to trimAl's `-automated1` column-filtering mode (Capella-Gutiérrez et al., 2009, *Bioinformatics* 25:1972; trimAl removes poorly-aligned columns from an MSA, and `-automated1` is the heuristic designed for downstream maximum-likelihood phylogeny on diverged sequences) to retain 749 columns of the original 1,521. This is unsurprising: the 73 cluster representatives are by construction the most structurally divergent proteins in the family from each other, and the deep nodes connecting these distant lineages are not strongly resolved by either AA or 3Di phylogenetics at this scale. The relationships within each structural class (as illustrated by the FS_001 analysis above) are more informative than the relationships between classes.

Q2: FS_001 AA tree (n=338)

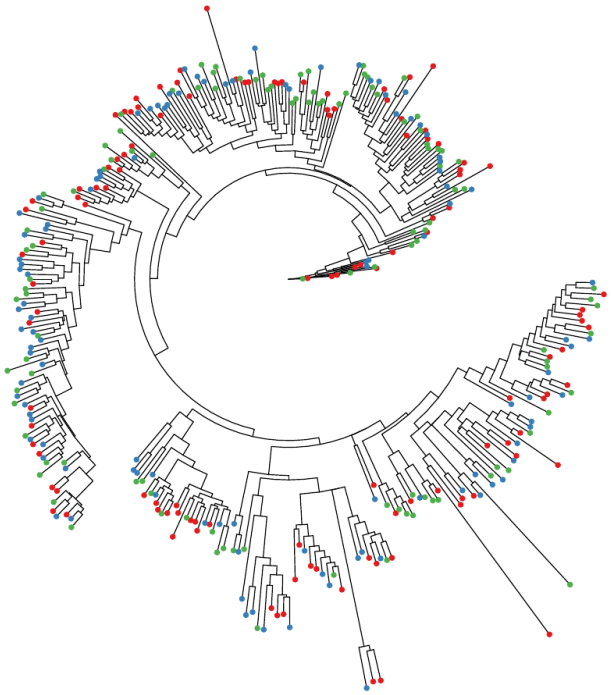


Figure 5: Maximum-likelihood phylogeny of 338 representative sequences from the dominant structural class FS_001 (118 from PvP01, 110 from PvW1, 110 from PvPAM), inferred with IQ-TREE 3 using VT+F+R6 and 1000 ultrafast bootstrap replicates. Tips are colored by genome of origin.

0.7 Methods

0.7.1 Data sources

Genome assemblies, protein FASTAs, and gene-level GFF annotations for all three reference genomes were downloaded from PlasmoDB release 68/71 (Amos et al., 2022, *Nucleic Acids Res.* 50:D898) at `https://plasmodb.org/common/downloads/Current_Release/{PvivaxP01,PvivaxPvW1,PvivaxPA`
The 3Di substitution matrix in NEXUS format was sourced from the jonasmehl1/structural_phylogenetics

repository (originally from Garg & Hochberg 2025, *Mol. Biol. Evol.* 42:msaf124). The Pfam HMM PF05795 ("Plasmodium_Vir") was fetched from Pfam-A.hmm (Mistry et al., 2021, *Nucleic Acids Res.* 49:D412) via `hmmfetch`.

0.7.2 VIR protein identification

For each genome, all annotated proteins in the PlasmoDB protein FASTA were searched with `hmmsearch` from HMMER 3.4 (Eddy, 2011, *PLoS Comput. Biol.* 7:e1002195) against PF05795. Proteins passing the default per-sequence threshold were considered VIR-family. This identified 931 / 748 / 985 functional VIR proteins in PvP01 / PvW1 / PvPAM respectively, summing to 2,664 across the three genomes.

0.7.3 Pseudogene characterization

Pseudogenes are absent from the protein FASTAs by construction (only translated coding sequences are released as proteins) so a separate workflow was needed to characterize them. For each genome:

1. **Identify pseudogene records.** All `pseudogenic_transcript` features were extracted from the PlasmoDB release 68 GFF along with their constituent exon coordinates (chromosome, start, end, strand).
2. **Extract nucleotide sequences.** Each pseudogenic transcript's exon sequence was extracted from the genome FASTA, exons concatenated in genomic-coordinate order, and reverse-complemented if encoded on the minus strand.
3. **Six-frame translation.** Each pseudogenic transcript nucleotide sequence was translated in all six reading frames (three forward, three reverse-complement) using the standard genetic code. Stop codons were translated as $*$ and retained in the resulting amino acid sequences (since pseudogenes frequently contain premature stops).
4. **HMM search.** All six-frame translations of length ≥50 amino acids were searched with `hmmsearch` against PF05795 at E ≤ 10^{-3}.
5. **Per-pseudogene call.** A pseudogenic transcript was classified as a VIR-family pseudogene if any of its six-frame translations produced a significant hit. The PvP01 result (75 VIR-family pseudogenes out of 185 total) is a strict subset of the 119 PvP01 pseudogenes that PlasmoDB independently product-labels as PIR/VIR, validating the HMM-based approach.

This workflow returns 75 VIR-family pseudogenes for PvP01 and 70 for PvPAM. PvW1's PlasmoDB release does not include any pseudogene annotations, so the same workflow cannot be applied; the corresponding cell in Table 1 is therefore "not determinable" rather than "0."

0.7.4 Structural clustering

The 2,664 VIR proteins were combined into a single FASTA and clustered with `foldseek easy-cluster` (Foldseek v10.941cd33; van Kempen et al., 2024, *Nat. Biotechnol.* 42:243) using `--prostt5-model` to encode the 3Di structural alphabet on-the-fly from the amino acid sequence (ProstT5 weights downloaded with `foldseek databases ProstT5`; Heinzinger et al., 2024, *NAR Genom. Bioinform.* 6:lqae150). Clustering parameters: `-c 0.8 -e 0.001 --alignment-type 2` (combined 3Di+AA alignment, 80% mutual coverage, E ≤ 10^{-3}). Output clusters were re-named FS_001 through FS_073 in decreasing size order. As a sequence-only baseline, the same 2,664

proteins were also clustered with `mmseqs easy-search` followed by Markov Clustering (MCL v22.282; van Dongen, 2008, *SIAM J. Matrix Anal. Appl.* 30:121) at inflation 2.0.

0.7.5 Spatial analysis

Gene coordinates were parsed from the `location=` field of the protein FASTA headers. The 14 largest contigs per genome were designated "main chromosomes" and used for spatial statistics to control for the differing subtelomeric fragmentation across the three assemblies. Subtelomeric enrichment was quantified as the fraction of main-chromosome VIRs lying within the outer 10% of their chromosome's length. Tandem clustering of same-cluster members was tested by computing, for each (cluster, genome) pair with ≥2 same-chromosome members, the mean distance from each VIR to the nearest same-cluster VIR on the same chromosome, compared against the distribution of the same statistic across 500 random shuffles of cluster labels within the genome. Cross-genome spatial-signature conservation was assessed by computing per-cluster summary statistics (number of main chromosomes occupied; median relative distance to chromosome end; fraction in outer 10%) and computing Spearman rank correlations between pairs of genomes over the 21 universally-shared clusters.

0.7.6 Phylogenetic inference

FoldMason v4 (Gilchrist et al., 2026, *Science*, doi:10.1126/science.ads6733) was used for multiple structural sequence alignment via `foldmason structuremsa` against the Foldseek database, producing aligned amino acid and 3Di sequences in lockstep. Within FS_001, a 50% sequence-identity subsample (338 sequences) was generated with `mmseqs easy-cluster --min-seq-id 0.5 -c 0.8` before alignment. Alignments were trimmed with trimAl 1.5 (Capella-Gutiérrez et al., 2009, *Bioinformatics* 25:1972) using `-gt 0.65` for FS_001 (paper-standard cutoff) and `-automated1` for the family-level set (where the per-class divergence is much higher). Maximum-likelihood trees were inferred with IQ-TREE 3.1.2 (Minh et al., 2020, *Mol. Biol. Evol.* 37:1530, describing the IQ-TREE 2 methodology) using ModelFinder (Kalyaanamoorthy et al., 2017, *Nat. Methods* 14:587) for AA model selection and the Garg & Hochberg 2025 substitution matrix (`-mdef 3DI.nexus -m 3DI+G`) for 3Di trees. Each tree was inferred with `-B 1000 --bnni --alrt 1000 -T 8 --safe --seed 42` (1000 ultrafast bootstrap replicates with NNI refinement, 1000 SH-aLRT replicates, 8 threads, fixed random seed). The genome-of-origin test for the FS_001 tree treated genome as a three-state discrete character and computed its parsimony score using `phangorn::parsimony`, compared against 1000 random permutations of the tip labels.

0.7.7 Cross-assembly orthology

Reciprocal best hits between the 349 PvW1 and 448 PvPAM members of FS_001 were computed with `mmseqs easy-rbh` (Steinegger & Söding, 2017, *Nat. Biotechnol.* 35:1026) using `--min-seq-id 0.3 -c 0.5 -e 1e-5`. A housekeeping ortholog backbone between the non-VIR proteomes of PvW1 and PvPAM was computed with the same tool at stricter thresholds (`--min-seq-id 0.5 -c 0.8 -e 1e-10`). For each FS_001 RBH pair, the 2 upstream and 2 downstream non-VIR neighbors of each member were identified by sorting all annotated proteins by chromosomal position; microsynteny score was defined as the number of those 4 PvW1 flanking neighbors whose housekeeping-RBH ortholog appeared among the 4 flanking neighbors of the PvPAM partner. Confidence classes were assigned as: high = microsynteny ≥ 3 AND identity ≥ 50% AND (not in Plan C Q2 tree OR sister leaves); medium = microsynteny ≥ 2 OR sister

leaves; low = otherwise. Sister-leaf relationships on the Q2 AA tree were computed by walking the tree topology with `Bio.Phylo` (Cock et al., 2009, *Bioinformatics* 25:1422) and assigning each leaf to its reciprocally-nearest leaf.

0.7.8 Computational environment

All analyses ran on macOS arm64 with conda-managed dependencies. Key tools and versions: HMMER 3.4, MMseqs2 18.8cc5c, Foldseek 10.941cd33, FoldMason 4.dd3c235, MCL 22.282, trimAl 1.5.rev1, IQ-TREE 3.1.2, R 4.4 with `ape`, `treeio`, `ggtree`, `phangorn`, Python 3.11 with `pandas`, `scipy`, `matplotlib`, `seaborn`, `biopython`.

0.8 Galaxy implementation

To translate this structure-aware phylogenomics workflow into a standard Galaxy pipeline, the analytical steps can be split into two classes based on their availability in the current Galaxy ecosystem (usegalaxy.org and the IUC ToolShed):

0.8.1 Standard ToolShed integrations

The sequence-based, standard phylogenomics components can be executed directly using official IUC-wrapped tools: 1. **Functional PIR Identification:** Executed using the standard `hmmsearch` tool (`toolshed.g2.bx.psu.edu/repos/iuc/hmmer_hmmsearch/hmmer_hmmsearch/3.4+galaxy0`) against custom-uploaded or pre-indexed HMM database files (`PF05795`). 2. **Subsampling and RBH Ortholog Candidate Generation:** Executed using MMseqs2 clustering and sequence-search tools (`toolshed.g2.bx.psu.edu/repos/iuc/mmseqs2_easy_linclust_clustering/mmseqs2_easy_linc`
3. **Phylogenetic Tree Construction:** Executed using IQ-TREE 2/3 (`toolshed.g2.bx.psu.edu/repos/iuc/iqtre`
which has built-in ModelFinder and supports the upload and use of custom substitution matrices (such as the Garg & Hochberg 2025 3Di substitution matrix).

0.8.2 Custom User-Defined Tool (UDT) requirements

The more specialized, structure-aware steps of the pipeline utilize emerging methods that are not yet packaged in the standard ToolShed. In a modern Galaxy instance, these can be incorporated natively as containerized **User-Defined Tools (UDTs)** without requiring administrative privileges: 1. **Foldseek Structural Clustering (v10.941cd33):** Requires a containerized UDT utilizing the official `foldseek` Docker image, loaded with the pre-compiled ProstT5 language model weights to encode amino acid sequences into 3Di alphabets on-the-fly and perform structural set-cover clustering. 2. **FoldMason Structural MSA (v4):** Requires a UDT wrapping the `foldmason` container to perform lockstep alignments of amino acids and 3Di structural sequences. 3. **Flanking Microsynteny Filtering:** Implemented via a lightweight Python-based UDT that takes the MMseqs2 RBH candidate pairs and cross-references their neighboring genes against the non-VIR housekeeping-RBH backbone. 4. **Pseudogene Rescue GFF Parser:** Implemented as a custom python tool that parses GFF3 annotations, extracts pseudogenic transcripts, translates them in all 6 reading frames, and streams the output to HMMER. 5. **Spatial permutation tests:** A custom tool executing nearest-neighbor distance label-shuffling tests to verify physical tandem cluster co-localization.

The complete tool execution ledger, input/output mappings, parameter sets, and the raw Python and R code blocks representing the custom-tool logic are fully documented in the standalone file `pipeline.md`. This blueprint allows a researcher to either execute the pipeline locally using a Conda environment or wrap the steps as a highly reproducible, sharing-enabled Galaxy workflow.

0.9 Summary

Three high-quality reference assemblies of *P. vivax* – PvP01 from Indonesia, PvW1 from Thailand, and PvPAM from Peru – support a coherent characterization of the PIR/VIR family that differs in several important respects from the picture available from the historical Salvador-I reference. The family contains approximately 750-1000 functional members per genome (with the exact number depending on assembly contiguity); these resolve into approximately ten major structural classes plus a long tail of rare variants; one structural class, FS_001, accounts for 46% of the entire family by itself; 97% of all VIR proteins fall into the 21 structural classes that are universally represented across the three continents; the family occupies all 14 main chromosomes in each assembly, concentrated at chromosome ends but interleaved by structural class rather than tandemly arrayed; the per-cluster spatial distribution is conserved across assemblies (Spearman $\rho \approx 0.91$ between any two genomes); and within the dominant structural class, the sequence diversity is high but not partitioned by geographic origin – consistent with a family whose diversification predates the modern *P. vivax* lineages and that continues to homogenize through subtelomeric recombination.

The orthology analysis quantifies how much of this family-level conservation translates into individual-gene conservation across the long-read assemblies: only about 6% of within-FS_001 reciprocal best hits between PvW1 and PvPAM survive a microsynteny verification step and qualify as high-confidence 1-to-1 orthologs. The remaining 94% of RBH pairs are best explained as paralog-best-hits, in which the closest match in the other genome is not the true ortholog but a related paralog from the same structural class. For methodological purposes this means that orthology in the VIR family cannot be inferred from sequence similarity alone, even with reciprocal best hits, and that confirmatory tests using microsynteny against a non-VIR ortholog backbone are necessary.

The natural next steps are: (i) extending the same pipeline to the Salvador-I reference, ideally after re-assembly with long reads, to bridge the historical A-L subfamily nomenclature to the FS_001-FS_073 structural classes; (ii) layering published transcriptome and proteome data over the structural classes, to ask whether stage-specific or condition-specific expression patterns are consistent with structural class membership; and (iii) repeating the cross-assembly characterization on additional *P. vivax* lineages from underrepresented regions (in particular Africa) to test whether the cross-continental conservation observed here generalizes to lineages not yet sampled.


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